

CADEN SOLLE

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EDUCATION

Sheboygan Area Lutheran High School

4.0 GPA, 36 ACT, Valedictorian

Liberty University

4.0 GPA

Graduating in May 2027 with a B.S. in Mechanical Engineering

PROFESSIONAL EXPERIENCE

Project Engineering Intern, Viking Masek

June 2026 - Present

- Supporting packaging system design by reviewing requirements, drawings, parts, and customer specifications.
- Performing sizing calculations, design checks, and equipment modification support.
- Assisting with risk documentation, troubleshooting, testing, and cross-functional coordination.
- Maintaining engineering records, vendor information, project data, and technical documentation.

Lead Payload Engineer, Liberty Rocketry

May 2025 - Present

- Lead a subteam, delegated tasks, and managed project timelines to ensure successful payload development.
- Designed a galvanometric laser engraver, including mirror steering mechanisms and custom Python code.
- Utilized CAD modeling and FEA to validate structural stability and optimize payload performance.

Undergraduate Research Assistant, Liberty University

Apr 2025 - Present

- Conducting high-strain-rate material testing with a Split-Hopkinson Pressure Bar system, including data acquisition, imaging, electron microscopy, and experimental protocol development.
- Assisting in the design and manufacturing of custom SHPB systems using GD&T to fit material testing needs.

Aerodynamics Engineer, Liberty Formula SAE Racing (35th of 150 teams in 2025)

Sep 2023 - Nov 2025

- Performed CFD analysis in ANSYS Fluent to enhance aerodynamic efficiency, achieving 14% less drag and 43% more downforce generation.
- Modeled, optimized, and manufactured numerous carbon fiber parts while balancing both weight and strength.
- Created complex assemblies using CAD software, ensuring full integration with chassis geometry.
- Designed and developed the first-ever rear wing for Liberty Formula Racing.

Mechanical Engineering Intern, Trinity Consulting and Design

May 2025 - Aug 2025

- Interpreted project specifications, technical drawings, and BIM documentation while using Bluebeam and Revit to create and analyze electronic security system designs.
- Conducted site surveys to assess project requirements and inform layout and design decisions.

Drone Photography Business Owner, New Perspective Aerial

Jun 2020 - Present

- Founded and operating a drone photography business for a client base of 10+.
- Managing customer relationships, creating invoices, and handling business documentation.
- Editing and enhancing multimedia content using Adobe Photoshop, Lightroom, and Premiere Pro.

Engineering Ethics Competition Participant, Lockheed Martin

Feb 2026

- Developed and defended solutions for the responsible integration of AI into critical infrastructure.

PROJECTS

Unmanned Aerial Vehicle (UAV)

- Engineered a custom UAV with 3D-printed parts to support customizability and quick repairs.
- Created a large-scale, comprehensive assembly to allow modularity.
- Currently refining a payload-deployment system to accurately hit a 10-foot wide target.

Freestyle Drone

- Designed and fabricated a high-performance freestyle drone capable of speeds over 100 mph.
- Configured and programmed the onboard flight controller using Betaflight.

TECHNICAL SKILLS

- Machines/Tools: Lathe, Laser Cutter, Waterjet, Multimeter, Measurement Equipment
- Software: SolidWorks, ANSYS Fluent, ANSYS Mechanical, Onshape, Revit, Microsoft Office, INAV, OpenRocket
- Programming Languages: Python, MATLAB, Java

CERTIFICATIONS

CSWA Electronic Security Employee Agile Project Management Part 107 UAS License Ham Radio License